

PARTIDO STATE UNIVERSITY
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PSU RDE AGENDA 2023-2028





Message from the University President

I congratulate the Office of the University Research and Development Extension and Knowledge Management (UREKM) for this very significant document produced and crafted in concise and handy form to guide us in the conceptualization, development and full implementation of research and extension programs and projects and their utilization to support the rural and urban communities that we serve in Partido District and beyond.

I fully recognize and deeply appreciate the commitments and efforts of our key stakeholders and partners in their meaningful participation during the conduct of the RDE agenda setting exercises. Indeed, their RDE needs, experiences, as well as personal insights that are truly representative of their sectors provided the necessary elements that made this work participatory and inclusive.

It is very heartening to note that the RDE agenda that we crafted as made concretized by this document encompasses the most salient objectives under the millennium sustainable development goals (MSDG) which provide the compass for developing nations like us to emerge out of the most pressing socioeconomic, cultural and ecological challenges that confront our society and communities in our times and in the future.

I am hoping that the meaningful and relevant contexts and directions of this University RDE agenda will not only guide us in developing and implementing meaningful RDE projects and programs for our stakeholders, and partners, but also inspire us to continuously demonstrate productive and engaged scholarships in the university to promote sustainable and resilient communities.

All the best!

RAUL G. BRADECINA, Ph.D.
SUC President III



Message from the Vice President for Research, Extension and Knowledge Management

Over years of operation, the RDE agenda have guided us to implement quality programs and projects attuned to the needs of our stakeholders and partners. The university's continued implementation of relevant RDE is now demonstrable in our policies and programs that support our stakeholders and partners.

This RDE agenda which was crafted through the participation of our stakeholders and partners once again demonstrate our commitment not only for relevance and quality but also for our resolve to be in solidarity with them in confronting our society's current development challenges such as climate change, food security, environmental degradation, rural social enterprise, poverty, gender equality, learning loss and public health promotion among others, that require new information, technologies and innovations

Let this RDE agenda document not only serve as a guide for developing quality and relevant research, development and extension programs and projects to be funded by the University for our researchers and community extension workers, but also as a blue print for sustaining the trajectory of growth in our service area.

Endearing our institution to our stakeholders and community is made possible through the implementation of quality and relevant RDE that advocate and support sustainable development. I believe, this is the most effective way to instill quality and relevance in our missions to capacitate people and transform communities through knowledge creation and community service.

Congratulations to everyone.

PROF. PATRICIA M. CANDELARIA
Vice President for Research, Extension and Knowledge Management

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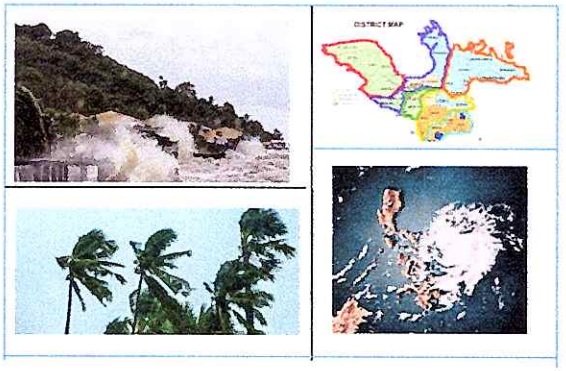
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Context of the RDE Environment

The geographic location of the University service area makes it vulnerable to the impacts of climate change. It is situated close to the Western Pacific makig it sensitive to the ocean-atmosphere dynamics which cause extreme weather events and hydrometeorological disasters such as extreme rain, drought, typhoons and flashfloods. The climate change induced variability in physico-chemical parameters of the coastal and marine environment enhances the virulence of water-borne microbes and pathogens threatening the health of the public residing in these coastal villages. Of the 10 municipalities of Partido District, the service area of PSU, 9 are coastal. These fishing villages as is common with other fishing villages in the developing world is characterized by poverty situation due to the common pool nature of the natural resource base of livelihood, and the remoteness of many fishing communities to the economic hubs in the province. The term "Partido" was derived from Spanish word meaning 'apartir" or separate referring to the geographical isolation of the district to the cengter of commerce and education in the province in the past. The constant threat of virulent climate change induced zoonotics to community health under an impoverished situation which constrain access and affordability of food highlight a situation of food insecurity making the people more vulnerable to health risks.





The threaths of poverty and food security provide heavy pressure on the fragile coastal and marine and upland ecosystems of the service area fuelling its constant degradation over time and compromising the sustainable production of ecosystem services and the livelihood of the majority of the resource-dependent communities. The natural-resource base dependency of majority of livelihood in the service area, and the persistence of low socioeconomic status of a lot of stakeholders in the countryside require the development of appropriate technologies that enhances agricultural and fisheries productivity and support the dynamism of small and medium scale enterprises.

The continued threat on the ecosystems and its degradation also exert undue pressure on the existence of indigenous communities which serve as the last bastion of Partido indigenous and pre colonial culture. The local traditional knowledge held by indigenous communities and the time-tested traditions of the people that are embedded in their tangible and intagible heritages are also impacted by the waves of modernization and consumerism that are now occurring in the rural communities which threaten the social capital and the social network of support that promotes resiliency to societal and climate changes. These situations necessitate the investments of RDE promoting TEK, local culture and history to pave the way towards establishing LEK and science driven, participatory governance in rural, indegenlous and resource-dependent communties in the service area.

Basis of the RDE Agenda

The RDE Agenda was guided by international and national development road maps particulalrly the United Nation Sustainable Development Goals and the Philippine Development Plan 2020-2028, and the Philippine Science, Technology, and Innovation Foresight: PAGTANAW 2050.

The 2030 Agenda for Sustainable Development provides a shared blueprint for peace and prosperity for people and the planet, now and into the future. At its heart are the 17 Sustainable Development Goals (SDGs), which are an urgent call for action by all countries in a global partnership. It recognizes that ending poverty and other deprivations must go hand-in-hand with strategies that improve health and education, reduce inequality, and spur economic growth – all while tackling climate change and working to preserve our oceans and forests (Figure 1).



Figure 1. United Nation Sustainable Development Goals

The Philippine Development Plan 2020-2028 is a plan for deep economic and social transformation to reinvigorate job creation and accelerate poverty reduction by steering the economy back on a high-growth path. It envisions that by 2028, Filipinos are envisioned to have long and healthy lives in livable communities with enough opportunities for high-quality lifelong learning for them to become healthy, smart, and innovative people (Figure 2).

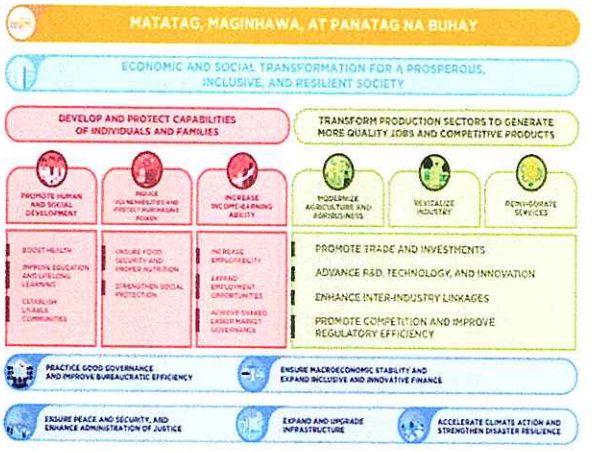
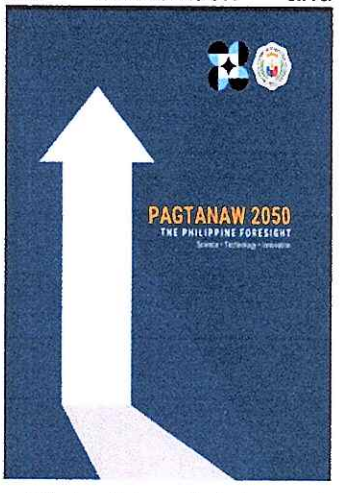


Figure 2. Strategy Framewrok of Philippine Development Plan 2023-2028

The Philippine Science, Technology, and Innovation Foresight is a compendium of STI megatrends, global and national societal goals, transdisciplinary, and interdisciplinary operational areas, and current and emerging technologies relevant to the nation's development. It delves into probable and significant drivers of change, and provides insights on the plausible development paths towards achieving the Filipino aspirations. At the core of this Foresight are 12 key operational areas, namely: blue economy; governance; business and trade; digital transformation and information and communications technology; science education and talent retention; food security and nutrition; health systems; energy; water; environment and climate change; shelter, transportation, other infrastructure; and space exploration.



Philippine Science, Technology, and Innovation Foresight: PAGTANAW 2050

The RDE Agenda also made use of key research and extension agenda developed and currently implemented by government and non government agencies as development road map as well. Among these are the National Harmonized Research Agenda (2017-2022), the National Unified Health Research Agenda (NUHRA) 2023-2028, the National Higher Education Research Agenda (NHERA) 2 , the Philippine Plan for Gender-Responsive Development, the SUCs R&D Consortiums Research Agenda, the Bicol Basic Research Agenda, and the Department of Education (DepEd) Region 5 Research Agenda.

The Harmonized National R&D Agenda (HNRDA) for 2022-2028 is aligned with national priorities and global advancements. ensure that science, technology, and innovation are geared towards and are utilized in areas of maximum economic and social benefit for the people. It covers renewable energy, sustainable and healthier alternative food products using local sources, solid waste management, and environmental quality concerns, among others (Figure 4).

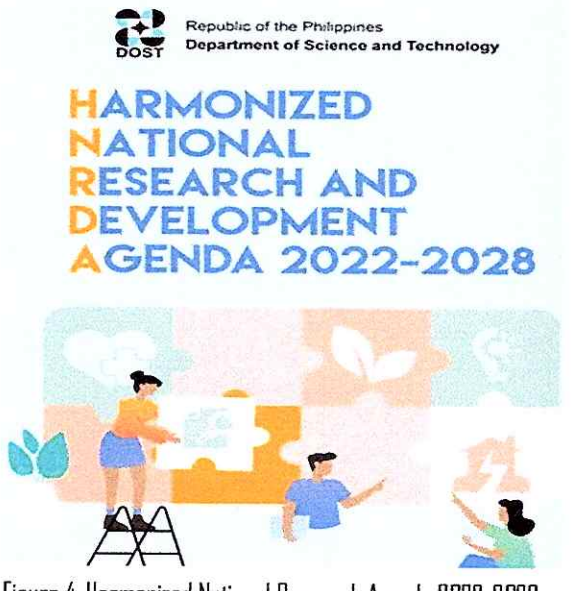


Figure 4. Harmonized National Research Agenda 2022-2028

The NUHRA 2023-2028 identified 170 health research priorities consolidated into 11 themes and sub-themes, namely: disease management, halal in health health security, emergency, and disaster risk management, health technology and innovation, health of vulnerable populations, health promotion health systems strengthening towards UHC, maternal, newborn, and child health, mental health nutrition and food security sexual and reproductive health (Figure 5).

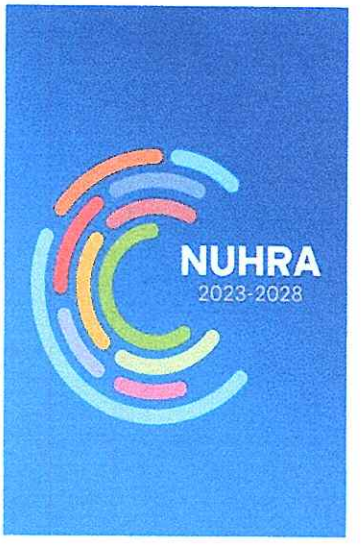


Figure 5. National Unified Health Research Agenda (NUHRA) 2023-

The National Higher Education Research Agenda (NHERA) 2 provides strategies and initiatives to develop research capacity and enhance research productivity in higher education institutions.

The NHERA also identifies priority areas for research and research-related programs in the next ten years.



Figure 6. National Higher Education Research Agenda (NHERA)

The Bicol Consortia on Agriculture. Industry and Energy, and Health Researches and Development agenda focus on the regional agriculture, industry and energy and health concerns. The agriculture research agenda cover aquatic, forestry, natural resources, and technology transfer and agriculture. The industry and energy research agenda cover product development and packaging, post-harvest equipment and machineries development, resiliency and sustainable initiatives, development of ICT software and programs, development of technologies for environmental management and disaster management, innovative technologies for transportation and infrastructure, food security, food innovation and halal program. The health research agenda focus on health governance, communicable diseases, maternal and child health, health workers and services delivery, non-communicable diseases, food security and health seeking behavior (Figure 7).

Bicol Consortium for Agriculture Aquatic and Natural Resources (BCAARRD) Agriculture, Aquatic and Natural Resources Priorities 2023-2028	Bicol Consortium for Industry, Energy and Emergent Technology R&D (BCIEERD) Agriculture, Aquatic and Natural Resources Priorities 2023-2028	Bicol Consortium for Health Research and Development
A. Agriculture Crops Livestock	A. Product Development and Packaging R&D activities that focus on the development and/or packaging of locally available natural resources in Bicol into food or non-food commodities	A. Health Governance
B. Aquatic	B. Postharvest Equipment and Machineries Development R&D projects that develop post-harvest equipment for locally-produced food or non-food commodities (i.e. fabrication and pilot testing of new technology/innovation of existing technology)	B. Communicable Diseases
C. Forestry and Natural Resources	C. Resiliency and Sustainability Initiatives Innovative technologies (framework tools/warning devices/systems) that would mitigate threats and impacts by (of) natural or man-made hazards to communities, business establishments and agricultural and aquatic resources)	C. Maternal and Child Health
D. Technology Transfer	D. Development of ICT / Software /Program Projects that concentrate on developing software or applications to upgrade services of government and MSMEs Smart cities focusing on urbanization challenges Disaster Risk Reduction Management and cyber security	D. Health Workforce E. Health Service Delivery F. Non-Communicable Diseases
	E. Development of Technologies For Environmental Management, Disaster Risk Management R&D projects and technology transfer along environmental conservation, pollution control and wastewater management	G. Food Scarcity and Security H. Health-Seeking Behavior
	F. Innovative Technology for Transportation and Infrastructure Bio-based building materials (preferably waste from agricultural sector) Recycled concrete as aggregate	I. Disaster Risk Reduction and Management J. Mental Health
	G. Food Security, Food Nutrition, Food Innovation and Halal Program	

Figure 7. RDE agenda of the agriculture, industry and energy and health research consortia in Bicol

The Bicol Basic Research agenda covers a total of 19 concerns namely, climate change and DRRM, vulnerable ecosystems, ethnography, traditional knowledge, living traditions and crafts, social enterprise, potential products/material science, defense and security, biodiversity conservation and ecological studies, ecotoxicology, plant genetics, limnology, bioactivity of plants and animal resources, pathology, alternative energy, parasitism and behavioral economics (Figure 8).

Bicol Basic Research Priorities
A. Climate Change and DRRM
B. Vulnerable Ecosystems
C. Ethnography
D. Traditional Knowledge
E. Living Traditions and Crafts
F. Social Enterprise
G. Potential Products/Material Science
H. Defense and Security (DaS - KTOP)
I. Biodiversity Conservation
J. Ecological Studies
K. Ecotoxicology
L. Plant Genetic Research
M. Biodiversity Conservation
N. Limnological Studies
O. Bioactivity Studies of Plants and Animals Resources
P. Pathology
Q. Alternative Energy
R. Parasitism
S. Behavioral Economics

Figure 8. Bicol RDE agenda for Basic Research

The Updated Gender Equality and Women's Empowerment (GEWE) Plan 2019-2025 supports and aligns with the remaining years of the Philippine Plan for Gender-Responsive Development (PPGD) 1995-2025. It contains strategic actions that: respond to the changing needs of women and girls because of the COVID-19 pandemic and move the country closer to the achievement of the PPGD's long-term vision of gender equality and women's empowerment (Figure 9)..

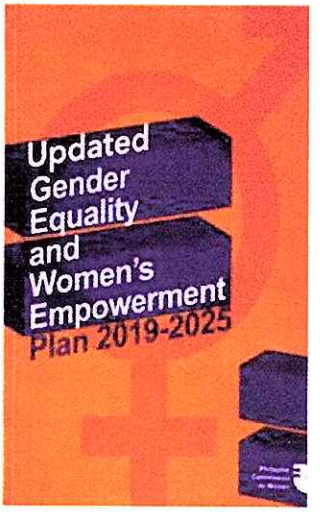


Figure 9. Updated Gender Equality and Women's Empowerment (GEWE) Plan 2019-2025

The Basic Education Research agenda makes known the research priorities of DepEd across four themes: teaching and learning, child protection, human resource development and governance and three cross cutting themes: gender and development, disaster risk reduction and management and inclusive education (Figure 9)

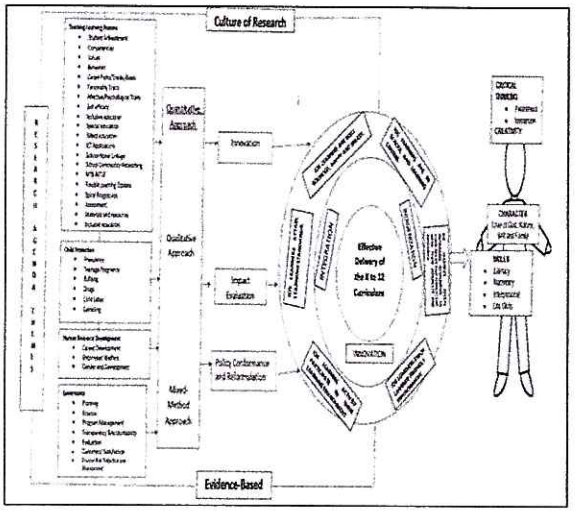


Figure 9. DepEd Basic Education research agenda

PSU recognizes the importance of the seventeen (17) sustainable development goals in reducing poverty, ending inequality, enhancing economic progress and conserving the natural resources. The university to offer researches on empowering micro, small and medium enterprises (SMEs), gender and development, educational reforms, responsive health systems, environmental protection and climate change mitigation and adaptation.

As a research academic institution, PSU adopted the research priority programs of the HNRDA reflecting them on its research agenda. PSU endeavors to produce researches on the areas of industry, energy, and emerging technology, health, clean energy, food and nutrition security, and on disaster risk reduction and management.

PSU upholds and supports the goals of the Philippine Development Plan 2020-2028.. The University believed that it is through research and development where Filipinos can attain a more comfortable and progressive life. Thus, the research thrust of the University is pushing toward the research and development that promote long and healthy lives in livable communities with enough opportunities for high-quality lifelong learning to become healthy, smart, and innovative people. Also, the PSU recognizes the importance of health-related researches in promoting public health. Thus, it adopted the different priority programs under the NUHRA as among the bases in developing the R&D agenda. The university also upholds the importance of promoting research needed for the advancement of higher education as well as for national development. It is with this reason that the four research agenda of the NHERA were used as a basis in crafting the University's R&D agenda for the year 2023- 2028.

PSU promotes the empowerment of women as well as persons with disability. Hence, the Gender Equality and Women's Empowerment (GEWE) Plan served as the bases in crafting the research agenda on gender and development. It highly recognizes the important role of research in regional development. Hence, on crafting its research agenda, the priority research programs of the .research consortia in Bicol were adopted and integrated in the agenda setting. The research priority programs of DOST research consortia in Bicol region focuses on competitive and innovative industry and services sector, competitive and sustainable agriculture and fisheries, accelerating social development , conservation and rehabilitation of the environment and natural resources and community promoting health and disaster reduction.

Partido State University acknowledges the salient role of basic education research in regional development. It is for this reason that the reformulated research agenda of the DepEd was integrated in the formulation of the University's research agenda. The University also supported and upholds researches on teaching and learning, child protection, human resource development as well as on good governance.

Method Used in R&D Agenda Setting

Each stakeholder involved during the parrticipatory agenda setting carries with them personal biases and knowledge which may or may not be entirely accurate and reflective of development issues. To lend structure to the information provided to the participants during the information extraction process, the PSU R&D research and development agenda adopted the Kingdon Model for agenda setting (1984).

The Kingdon Multiple Streams Model on Agenda Setting was integrated in the process. The model was based on societal predisposition in determining problem stream, policy stream and political stream before the actual formulation of the University's research agenda. Societal predisposition focuses on the context of international, national, regional and local researches. For this reason, the United Nation Sustainable Development Goals, the Philippine Development Plan 2020-2028, the Philippine Science, Technology, and Innovation Foresight: PAGTANAW 2050, the National Harmonized Research Agenda (2017-2022), the National Unified Health Research Agenda (NUHRA) 2023-2028, the National Higher Education Research Agenda (NHERA) 2 , the Philippine Plan for Gender-Responsive Development, the SUCs R&D Consortiums Research Agenda, the Bicol Basic Research Agenda, as well as the Department of Education (DepEd) Region 5 Research Agenda were incorporated on the agenda.

Figure 10 depicts the Kingdon Multiple Stream Model on agenda setting.

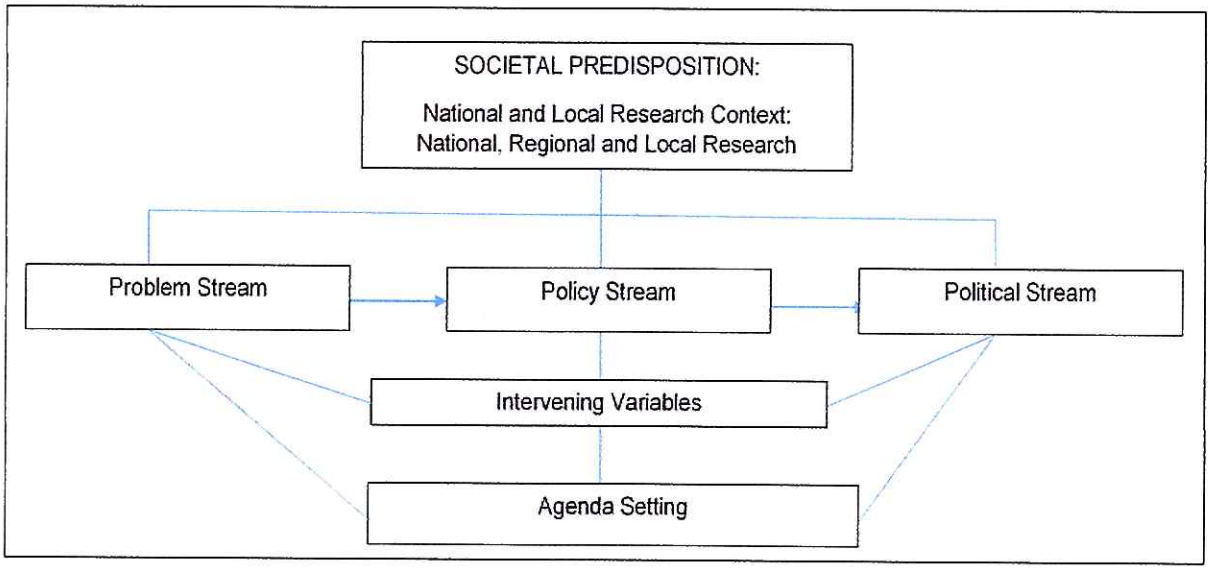


Figure 10. The Kingdon Multiple Stream Model on Agenda Setting

The PSU RDE Agenda

The PSU research, development and extension contextually and purposively work with all the international, national, regional and local initiatives and directions mentioned above to deepen understanding of ourselves as a people and as a nation, discover parctical, evidence-based answers that can address real-worl social, economic and environmental chaalleneges of familes and communties. In this context, the RDE agenda will focus on themes that are based on the key development, research and innovation documents used as inputs to the RDE agenda setting namely:

I. Food Production and Security

II. Health Systems

III. Environment, Disaster Risk Reduction, Climate-Change and Energy

IV. Education in Science, Technology, Agri-fishery and Mathematics) STEAM and Culture

V. Terrestrial and Marine Resources, Economy, Biodiversity and Conservation

VI. Smart Analytics and Engineering Innovation

VII. Gender and Development

I. Food Production and Security

This theme serves the desired national development goal of increased agricultural and fisheries peoductivity as well as the availability and accessibility of safe and nutritious food that meet people' dietary needs.

II. Health Systems

This theme is aimed at improving evidence-based knowledge and understanding on how to optimize health service delivery using a systems approach. It is cocnerned on how social factors, governance, workforce, finncing systems, products and technologies, organizational structrues and processes, persoal behaviors and information-seeking actions affect access to halth care, the quality and cost of health care, and ultimately our health and well-being. Hence th end-view is to identifiy the most effective ways to organized, manage, fiance and devlievr high quality care, reduce medical erros; and improve patient safety.

III. Environment, Disaster Risk Reduction, Climate-Change and Energy

This theme si aimed at improving knowledge and underatanding of the global environment and climate change, generating systems and public consciousness to conserve local environments, and developing locally apprprite technologies,systems and tools and procedures in preparation for natural disasters.

IV. Education for Science, Technology, Agri-fishery and Mathematics) STEAM and Culture

This theme is aimed at generating innovations in undergraduate education in Science, Technology, Engineering, Agriculture-Fisheries,and Mathematics (STEAM). It will also generate information on th4 whole complex of distinctive spiritual, intellectual, emotional and material features that characterise the Bikolano and Partido societies as social group. It includes the arts and letters as well as human modes of life, value systems, creativity, knowledge systems, traditions and beliefs.

V. Terrestrial and Marine Resources, Economy, Biodiversity and Conservation

This theme is aimed at generating sustainable development strategies for activities based on the use of land and ocean resources. It also aimed at understanding behavior of resource users in sustainable resource governance.

VI. Smart Analytics and Engineering Innovation

This theme is aimed at fostering and enabling research that can help data users such as small to medium business enterprises, industry, local governments, legislators, policy makers, teachers, students, individuals and families, and other stakeholders harness the potential of data, in varying levels of complexity, in order to make informed decisions, innovations such as smart predictive informatics tools are needed to help make sense of ever increasing configurations of data, and thereby aid in day-to-day productivity as well as support macro goals of economic competitiveness.

VII. Gender and Development

This theme is aimed at generating information under a feminist approach to understanding the disparate impact that economic development, resource utilization and resource governance have on people based upon their location, gender, class background, and other socio-political identities. It shall also implement studies under an economic approach to community's development in quantitative terms such as job creation, and employment to improve 'economic wellbeing' and the subsequent quality of life. In terms of economic development, quality of life is defined as access to necessary rights and resources including but not limited to quality education, medical facilities, affordable housing, clean environments, and low crime rate. It is also aimed at understanding how multifaceted these issues are in the entangled context of culture, government, and globalization.







Research Priorities and Studies

I. Food Production and Security

Priority Area	Research Topics
A. Agriculture and Fisheries Farming Systems and Production Technologies	1. Aquasilviculture 2. Hatchery and Mariculture of Aquatic Species 3. Aquaponics 4. Hydroponic system 5. Crop and Animal Production 6. Agricultural Technology Development 7. New/Endemic species for culture 8. Mechanization and Automated Systems 9. Good Agricultural Practices 10. Diversified farming system 11. Commodity Roadmap for Aquaculture, Agriculture in Partido Area 12. Profiling of existing aquaculture facilities and production in Partido 13. Verification Studies of Local Marine Species for Aquaculture in Partido Area 14. Farming Systems and Calendar System for Planting and Harvesting Crops in Bicol 15. Farming Systems of Traditional Root Crops in Upland and Wetland Areas (*Galyang, Gabi, Arorot*) 16. Ichthyoplankton studies 17. Aquaculture farming system and sea ranching for abalone 18. Propagation and production of traditional herbs and spices (*Luya*, Turmeric, *Batuan, Lubas*, etc) 19. Digitization of agriculture and fisheries farming systems
B. Agribusiness and Marketing	1. Community-managed enterprises 2. Financial/economic viability evaluation of agricultural technologies 3. Market Research
C. Value Chain	1. Priority commodities
D. Food Governance	1. Policy research 2. Ensuring sustainability of food sources 3. Technology needs assessment for micro enterprises (food products) 4. Operational, policy and institutional models in ensuring food and nutrition security 5. Transdisciplinary approaches in building sustainability of food systems 6. Livelihood support system
E. Food Safety	1. Foodborne diseases
F. Social Protection	1. Food security and peace building 2. Support to farmers organization (Organization and Registration)
G. Post-Harvest Handling, Processing and New Product Development	1. Product development and packaging 2. Pilot testing of technologies 3. Rollout/commercialization 4. Nutritional analysis of bamboo shoots (Lambo) 5. Value-adding for Abalone 6. Small-scale Enterprise Development for Seaweeds in Garchitorena, Siruma and Caramoan 7. Abalone processing as small-scale enterprise development in Lahuy Group of Islands, Caramoan 8. Development of sea-based product facilities 9. Verification study and establishment of seaweeds drying facilities 10. Traditional fruit-bearing trees product development (Rimas/Kulo, Gumihan, Bulala, Kalpi) SALT leading to watershed development in Siruma 11. Pili product development
H. Biotechnology application for food production	1. Fish health management 2. Application of Genomics 3. Disease diagnostics and vaccines
I. Production and Management Decision Support System	1. Technology Business Incubation (TBI) 2. Community-based management of natural resources 3. LGU egovernance system
J. Mechanization and engineering technology	

II. Health Systems

Priority Area		Research Topics
A. Service Delivery		
I. Medical Services		
	A. Health and Nutrition	1. Assessment of maternal and child health 2. Health seeking behavior in disadvantaged areas in Partido 3. Food supplement and functional food development 4. Natural products in drug discovery and development 5. Association studies on health and SDG 2 (Zero Hunger)
	B. Water, sanitation and hygiene (WASH)	1. Assessment of water supply, sanitation, and hygiene practices 2. Water quality monitoring in level 1 water sources 3. Technologies and strategies to improve water quality and reduce water-related diseases 4. Design of low-cost excreta disposal systems
	C. Zoonotic Diseases	1. Detection and prevention of emerging zoonotic diseases 2. Screening for antibiotic resistant microorganisms in the environment
	D. Health and Environment Nexus Governance	1. Assessing the mental health and well-being of students and employees 2. Links between environment, nutrition and mental health 3. Impact of climate change in the social and environmental determinants of health 4. Detection and monitoring of microbial and chemical contaminants in food' 5. Studies on environmental epidemiology 6. Linking governance mechanisms to health outcomes
	E. Sexual and Reproductive Health	1. Assessment of sexual and reproductive health knowledge and awareness in adolescents 2. Emerging issues in adolescent sexual and reproductive healthcare 3. Morbidity and mortality rates concerning sexual and reproductive healthcare
2. Health Promotion		1. IEC 2. Organization 3. Mobilization 4. End goal: Behavioral Change
3. Local Epidemiological Surveillance		1. Structure, Organization, Planning 2. People: Function, Training
4. DRRM- H		1. WASH 2. Health and Nutrition 3. Psychosocial or Mental Health 4. General Medical Services
B. Governance and Leadership		1. LGU and Policy
C. Health Information Technology		1. Electronic Medical Record 2. Tele-Medicine a. Tel-Consultation b. Tele-Mentoring 3. i-clinic system
D. Health Workforce		1. Hospitals 2. Primary Health Care Facilities
E. Access to Essential Medicine		1. Procurement 2. Supply Chain Management
F. Financing		1. Philhealth Health Insurance 2. Grants and Loans 3. Out-of-Pocket Spending 4. Source of Revenues and Funding

III. Environment, Disaster Risk Reduction, Climate-Change and Energy

	Priority Area	Research Topics
A.	Solid Waste and Plastics	1. Research and product development of waste 2. Awareness to community 3. Conduct assessment on the implementation of the national gov. policies
B.	Resilient socio-ecological communities	1. Disaster recovery and resilience bldg.
C.	Agri-tourism and eco-tourism and livelihood	1. Developing AgriTourism and EcoTourism in Partido District a. Assessment of Mangrove areas intended for eco-tourism parks.
D.	Futures Thinking Initiatives	1. Coastal Resource Management Biodiversity studies and biodiversity conservation

IV. Education for Science, Technology, Agri-fishery and Mathematics) STEAM and Culture

	Priority Area	Research Topics
A.	Bicol History and Culture	1. Origin of Indigenous Peoples in Partido 2. Documentation of IP's Best Practices (Land conservation, healing, food production, farming, etc.) 3. Documentation of Oral Practices and Histories/Linguistics of IPs 4. Documentation of Partido indigenous knowledge, culture, and heritage (10 Elements of Indigeneity) 5. Documentation of Extant arts, music, and language Instructional Materials Development on Oral Practices and Indigenous Knowledge for Reading and Mathematics Instruction 6. Indigenous Peoples & Protected Areas: Rights Principles and Practice 7. Impact of the Pandemic on Indigenous Peoples
B.	Customs and Traditions (Language & Literature)	1. Partido Studies on the Spanish Era 2. Partido Studies on the American and Japanese Era 3. Language Literacy Evolution 4. Instructional Materials Contextualization on Local Customs and Traditions for Reading and Mathematics Instruction
C.	Literacy and Numeracy	1. Assessment of Reading Readiness and Reading Competencies across Levels in Partido Area 2. Assessment of Listening/Reading Comprehension 3. Assessment of Mathematics Readiness and Numeracy Competencies across Levels in Partido Area 4. Reading and Mathematics Interventions across Levels in Partido 5. Investigating the Effectiveness of Reading and Mathematics Interventions across Levels in Partido 6. Factors affecting Reading and Mathematics Interventions (students' intrinsic motivations, attitudes, teachers' professional expertise, infrastructures, administrative support, funding, etc) 7. Disaster Risk Management in Early Childhood Education 8. Local Policy Research on Reading and Numeracy
D.	STEAM Education and Curriculum	1. Enrichment of STEAM Education and Curriculum a. Development of Multidisciplinary Instructional Materials for STEAM b. Integrating Contextualized

		Materials in STEAM Curriculum 2. Tracer Studies on STEAM Education Effectiveness IP Education in STEAM 3. Emerging Interventions/Innovations to support the 8-week Curriculum Implementation
E.	Gender and Economic Development	1. Investigating Gender Roles in the Economic Development in Partido 2. Inclusive Education in Partido (Mainstreaming of Learners with Special Needs) 3. Equal Opportunities Principle 4. Effect of Policies on IP Culture (ex. RA 11596- Anti-child Marriage Law
F.	Digitization and Artificial Intelligence	1. Automation / Digitization of Reading Comprehension Interventions across Levels 2. Utilization of Instructional Materials from the Learning Resources (LR) Portal 3. Development of Portal for Learning Resources

V. Terrestrial and Marine Resources, Economy, Biodiversity and Conservation

	Priority Area	Research Topics
A.	Forest, Marine and Aquatic Habitat Management	1. Forest Tree Improvement (provenance, progeny, clonal) of economically important species in Bicol Region a. Baseline data for the Economically Important Tree Species in Bicol Region 2. Development of seedling production protocols (coastal and forest species). a. Technical Bulletins from DENR b. Native and Endemic Germplasms (seeds, tissues and DNAs) from Protected Areas or Key Biodiversity Areas c. Seedlings that is adaptable for coastal areas and may be source of food. 3. Carrying Capacity of tourism areas among Partido District of in Bicol Region. 4. Determination of Key Biodiversity Areas in Partido District. 5. Plastics contamination in the marine environment....evaluation and measurement of the sources, distribution, fate, and impacts of marine plastics... a. Studies on microplastics
B.	Biodiversity Conservation, Assessment and Restoration	1. Biodiversity assessment of flora, fauna, microbial in selected KBAs, PAs, and Mountains 2. Phenological study of selected native and endemic tree species in Mt. Isarog Natural Park, CamSur 3. The use of novel or little-used approaches to document biodiversity, especially the less studied biodiversity-rich regions or habitats a. Evidence-based approach or Citizen Science approach b. Research on sustainability of potable water and agri use from Mt. Isarog
C.	Natural Resources Economy	1. Valuation studies of various ecosystem services (terrestrial and aquatic ecosystems) a. Resource Valuation studies in PAs (prioritization)
E.	Gender and Development	1. A value-chain based conservation and management assessment of Capacity of LGU personnel.

		2. Population density of Hornbills in Caramoan National Park (Species-specific protection and conservation) 3. Pawikan Conservation and Protection (Species-specific protection and conservation) 4. Taxonomic study of flora in Protected Areas (Species-specific protection and conservation) 5. Involvement of PSU in different activities of DENR and other stakeholders. 6. Community Education and Public Awareness (CEPA) re: Biodiversity Conservation 7. Initiatives: Study to Publish Initiative, encourage undergrads and graduate students to publish their studies.

VI. Smart Analytics and Engineering Innovation

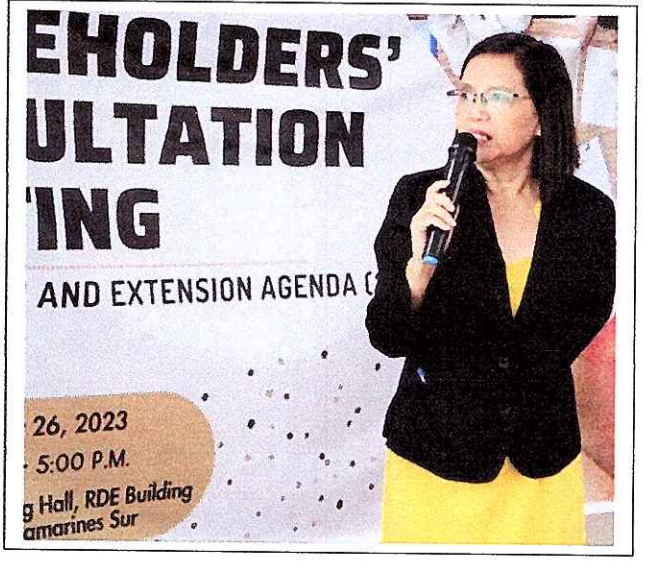
	Priority Area		Research Topics
A.	Information and Communication Technology	1.	SMART and Sustainable City/ Cloud Computing a. Tech-Startup (ICT-based solution for community-based problems) b. Digitalization of local government unit/ academe/business/MSMEs transactions (RA11032 -ease of doing business) Ex. online booking/ fin-tech/ e-Payments, delivery, logistics, education, health, purchasing/marketing, travel) c. Solid-waste Management Systems d. Stakeholders: -LGUs, -Industries, -MSEs, -Academe, -NGAs
B.	Decision Support System	2.	SMART Farming a. agri-tourism b. fish diseases c. identification of species d. harvesting (timeframe)
C.	Digital Innovations in Engineering	a.	Digital Innovation in Engineering a. Climate change resilient infrastructures
		b.	SMART Farming a. renewable energy utilization (solar energy and hydropower) b. rainwater harvesting technology
D.	Water Resources Engineering and Technology	c.	Water Supply System Engineering for Partido Area a. Development of Water Resources and Hydraulic Models
E.	Solid Waste Management	d.	Solid Waste Management (Minimization, Diversion / Conversion into Energy towards the Adoption of Circular Economy concept; IEC to raise the awareness of the community)
F.	Business Analytics	1.	Digital Technology as Strategy for Economic Inclusion of Rural Entrepreneurs a. Business Trend Analysis b. Supply & Demand Assessment c. Value Chain Analysis d. Digitalization of Business Operations (Inventory System, ePayments, Digital Marketing-Ecommerce Platforms etc.)
G.	Office Digitization	1.	Digitization of Records Management in Private and Public Offices

		a.	Cloud-based database (easier access of information, anytime, anywhere)
H.	GREEN & BLUE Economy	1. 2.	SMART agriculture SMART economy
I.	Mechanization and engineering technology		

VII. Gender and Development

	Priority Area		Research Topics
A.	Gendered Vulnerabilities in Disaster Situations	1.	Identify the specific gendered vulnerabilities and risks experienced in different types of disasters (look for topics to study like decision making process, empowerment, livelihood, governance, perceptions and behavior, policies
B.	Economic Contributions of Men and Women in agricultural production	1. 2. 3.	Identify the extent of men and women's participation in crop, livestock and poultry, and fisheries processing, marketing and related activities Gender-based framework of VCA Identify commodities contributing to economics, participation of women to ....
C.	Economic Contributions of Men and Women in Fisheries	1. 2.	Identify the extent of men and women's participation in fisheries, including their roles in fishing, processing, marketing, and related activities Gender contributions or Roles in fisheries, tuna, etc.
D.	Equal gender-based health, sanitation and reproductive rights	1.	Investigating men and women's access to healthcare services, sanitation, family planning, and reproductive rights, with a focus on marginalized communities
		2.	Participation to health care services, governance on health, perceptions and behavior on health, Engendering development programs, (health related),
E.	Gender-Based Violence	1.	Research into the incidence, prevalence and causes of gender-based violence
F.	Political Participation	1.	Analyzing the impact of policies promoting gender equality in politics, and the access of women in governance
G.	Digitization for men and women	1.	Analyze the extent of access and literacy of men and women in digital technology





Consistency of RDE Agenda with PSU Vision and Development Framework

The RDE agenda themes and priorities enhance and synergiize the quality of instructions in the various academic programs offered by the university to develop graduates who will fill the need for competitive manpower for today's agri-fisheries, services and industries and professionals who will lead the transfomation of rural communties. The RDE agenda themes which were guided by key development docuements and research directions and adopting the relevant plaforms for research, innovation and extension in higher education coherently address the interlinked social, ecological and economic pillars of the service area supporting the vision of PSU to become a catalyst of sustianable development in Partido and beyond (Figure 12).

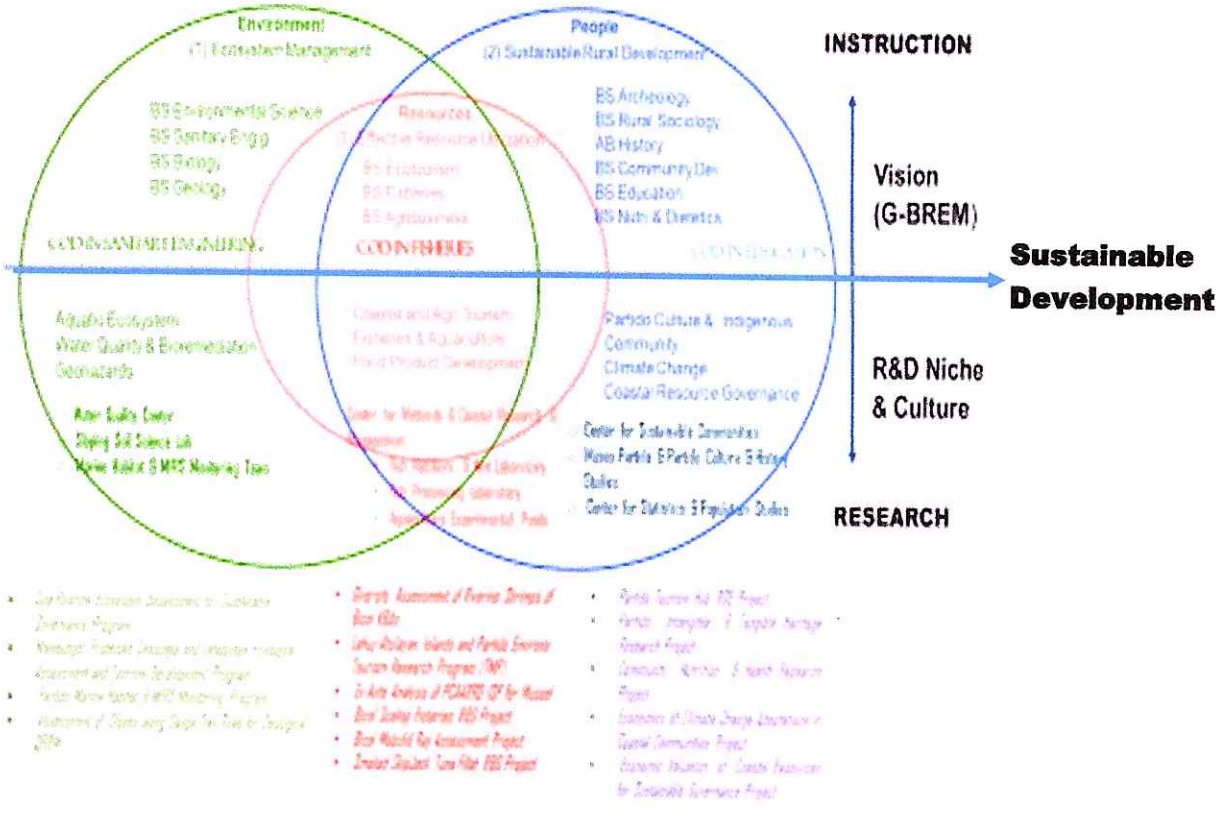
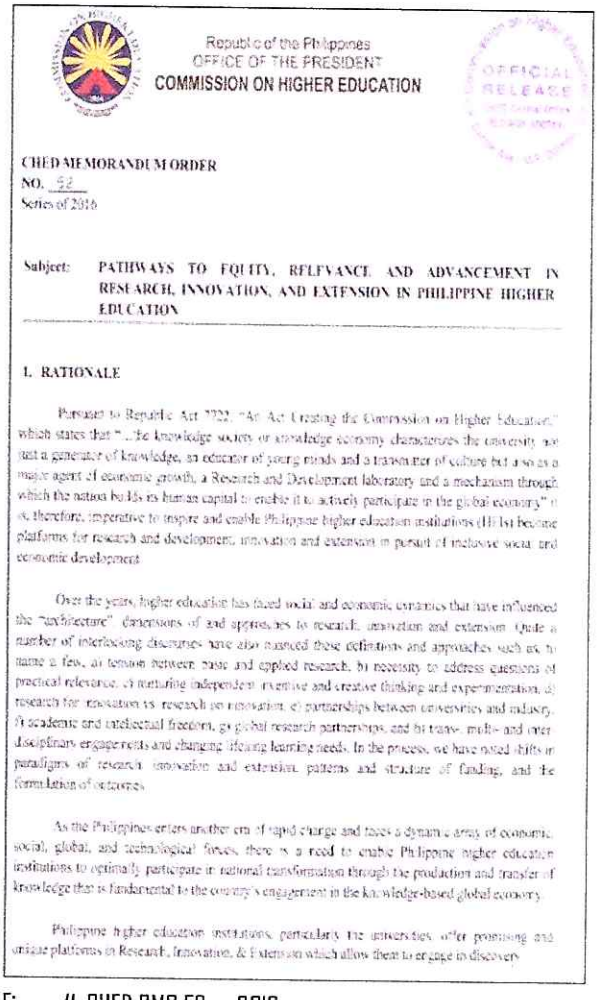


Figure 12. PSU sustainable development framework for Partido service area and adjacent communities

Alignment of RDE Agenda with CHED Platforms on Research, Innovation and Extension

To keep the University RDE in alignment with CHED Memorandum Order Number 52, series of 2016, which pursue a pathway for equity, relevance and advancement in research, innovation and extension in Pilippine Higher Education, this RDE agenda adopted most of the research and innovation platforms in its key thematic research areas.

Through this RDE agenda the University shares CHED's articulation of the strategic thrusts and pathways that can feasibly address the dynmaics in higher education and the development challenges of the times in order to fully actualize the potentials of PSU as platforms for knowledge production and advancement,s, engines of development through responsive and relevant research programs, and producers of multi-spepcialists, creators, problem-solvers, colllbaorators, inventors, thinkers and innovators who can examine phenomena, explore new frontiers and bring from multidisciplinary and interdisicplinay lenses (Figure 11)..



Republic of the Philippines
OFFICE OF THE PRESIDENT
COMMISSION ON HIGHER EDUCATION

OFFICIAL RELEASE

CHED MEMORANDUM ORDER
NO. 52
Series of 2016

Subject: PATHWAYS TO EQUITY, RELEVANCE AND ADVANCEMENT IN RESEARCH, INNOVATION, AND EXTENSION IN PHILIPPINE HIGHER EDUCATION

I. RATIONALE

Pursuant to Republic Act 7722, "An Act Creating the Commission on Higher Education," which states that "...the knowledge society or knowledge economy characterizes the university not just a generator of knowledge, an educator of young minds and a transmitter of culture but also as a major agent of economic growth, a Research and Development laboratory and a mechanism through which the nation builds its human capital to enable it to actively participate in the global economy" it is, therefore, imperative to inspire and enable Philippine higher education institutions (HEIs) become platforms for research and development, innovation and extension in pursuit of inclusive social and economic development.

Over the years, higher education has faced social and economic dynamics that have influenced the "architecture" dimensions of and approaches to research, innovation and extension. Quite a number of interlocking discourses have also nuanced these definitions and approaches such as, to name a few, a) tension between basic and applied research, b) necessity to address questions of practical relevance, c) nurturing independent inventive and creative thinking and experimentation, d) research for innovation vs research on innovation, e) partnerships between universities and industry, f) academic and intellectual freedom, g) global research partnerships, and h) trans-, multi- and inter-disciplinary engagements and changing lifelong learning needs. In the process, we have noted shifts in paradigms of research, innovation and extension, patterns and structure of funding, and the formulation of outcomes.

As the Philippines enters another era of rapid change and faces a dynamic array of economic, social, global, and technological forces, there is a need to enable Philippine higher education institutions to optimally participate in national transformation through the production and transfer of knowledge that is fundamental to the country's engagement in the knowledge-based global economy.

Philippine higher education institutions, particularly the universities, offer promising and unique platforms in Research, Innovation, & Extension which allow them to engage in discovery

Figure 11. CHED CMO 52, s. 2016



Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

LIST OF ATTENDEES
PSU Research and Extension Agenda (2023-2028) Setting
September 22, 2023 8:00 AM – 5:00 PM
RDE Training Hall, PSU Goa

1. Dr. Raul G. Bradecina
   SUC President III
2. Prof. Patricia M. Candelaria
   VP for Research, Extension and Knowledge Management
3. Ms. Luisa M. Lanciso
   Research and Development Director
4. Ms. Rosemarie A. Tabardillo
   Extension and Community Services Director
5. Dr. Mariel Estrella
   Knowledge Management Director
6. Mr. Michael Vale
   GAD Focal Person
7. Dr. Cristina P. Lim
   Center for Sustainable Communities Director / Sagñay Campus Dean
8. Dr. Danilo M. Gerona
   Center for Partido Studies Director
9. Ms. Maria Cristina G. Gumba
   Coastal Wetland and Ecosystems Center Director
10. Ms. Yumi Vivien De Luna
    ParSU Statistics Center Director
11. Dr. Joni Neil Capucao
    CAS Dean
12. Dr. Rina Abner-Puerta
    CBM Dean
13. Dr. Joan Monforte-Bedes
    COED Dean
14. Dr. Bernadette Gumba
    SGS Dean
15. Dr. Michael Clores
    Caramoan Campus Dean
16. Dr. Shirlene Esplana
    Lagonoy Campus Dean
17. Dr. Ricky Laureta
    San Jose Campus Dean
18. Dr. Romeo Morcilla
    Tinambac Campus Dean
19. Leomark Kevin D. Baldoza
    Federation President
20. John Kevin R. Alvaran
    Federation Vice President for Internal Affairs
21. Joshua P. Tenorio
    Federation Vice President for External Affairs
22. Ms. Liezel Atole
    Instructor I, CAS
23. Mr. Francis Ignatius Nieva
    Instructor I, CAS
24. Mr. Nicolas Pura
    Assistant Professor I, CAS
25. Mr. Salvador Briones
    Assistant Professor IV, CAS
26. Ms. Hazel Jean Pesino
    Assistant Professor III, CAS
27. Ms. Czarina Nicole Macatingrao
    Instructor I, CAS
28. Mr. Emmanuel Onsay
    Instructor I, CBM
29. Ms. Rhesty Biares
    Assistant Professor IV, CBM
30. Ms. Menchi Del Castillo-Crispino
    Instructor I, CBM
31. Ms. Kris Sarmiento
    Assistant Professor III, CBM
32. Mr. Jose Fernandez
    Associate Professor II, CET
33. Engr. Raymundo Romero
    Professor VI, CET
34. Ms. Ana Vargas
    Associate Professor IV, CET
35. Mr. Sherwin Pasasala
    Assistant Professor II, CET
36. Mr. Raymond Estrada
    Instructor I, COED
37. Ms. Shiela Amoroso
    Associate Professor I, COED
38. Ms. Josenia Penino
    Assistant Professor III, COED
39. Dr. Nerisa Paladan
    Associate Professor II, SGS
40. Dr. Wendy Solomo
    Associate Professor I, SGS
41. Dr. Abelardo Belleza
    Assistant Professor III, SGS

42. Ms. Diana Pesimo
    Instructor I, Caramoan Campus
43. Mr. Clarson De Vera
    Assistant Professor IV, Lagonoy Campus
44. Ms. Jeena Sandra De Hitta
    Instructor I, Lagonoy Campus
45. Ms. Margarita Lanciso
    Associate Professor I, Lagonoy Campus
46. Ms. Maria France de Vergara
    Instructor I, Sagñay Campus
47. Mr. Ronnel Dioneda Jr.
    Instructor I, Sagñay Campus
48. Mr. Ramel Earl Madrideo
    Instructor I, Sagñay Campus
49. Mr. Raul Burce
    Instructor I, Sagñay Campus
50. Ms. Princess Mae Lara
    Instructor I, Sagñay Campus
51. Dr. Jenny Lorio
    Associate Professor I, Salogon Campus
52. Dr. Gina Alcoriza
    Assistant Professor III, San Jose Campus
53. Engr. Kier Gasga
    Instructor I, San Jose Campus
54. Ms. Divina Chica
    Instructor I, Tinambac Campus
55. Mr. Ceazar Ian Pesimo
    Instructor I, Tinambac Campus

Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

LIST OF ATTENDEES
Stakeholders' Consultation Meeting
PSU Research and Extension Agenda (2023-2028)
September 26, 2023 8:00 AM – 5:00 PM
RDE Training Hall, PSU Goa

Thematic Area 1. Food Production and Security

1. Joel B. Balasta
   Municipal Agriculturist,
   LGU Siruma
2. Atty. Nilo B. Consuelo
   Chief Aquaculturist,
   Bureau of Fisheries and Aquatic Resources (BFAR) Region 5
3. Lea M. Beltran
   Municipal Agriculturist,
   LGU Tigaon
4. Eme V. Alvarez
   Treasurer,
   Pawican Kasmay Association (PAKASA)
5. Lyssa Alvarez
   Member,
   Pawican Kasmay Association (PAKASA)
6. Nida A. Hagos
   Acting Planning Officer / DMO I,
   Department of Agriculture-Agricultural Training Institute (DA-ATI) Regional Training Center Region 5
7. Andrew V. Briones
   Municipal Fishery Coordinator,
   LGU Garchitorena

Thematic Area 2. Health Systems

1. Rey J. Millena, MD
   Provincial DOH Officer

Thematic Area 3. Environment, Disaster Risk Reduction, Climate-Change and Energy

1. John Ray Cambaling
   SEMS,LGU Tinambac
2. Edwin San Juan
   OIC-AHRD/BO III,
   Partido Development Administration
3. Jaime A. Coralde
   Chairman,
   Sagñay Rigaton Agriculture Cooperative (SARIGACO)
4. Kenneth P. Zamudio
   President,
   Rotary Club of Goa Partido
5. Alvin B. Burac
   OIC – Fish Capture,
   Bureau of Fisheries and Aquatic Resources (BFAR) Region 5
6. Geraldine Consuelo
   Technical Staff – LHD,
   Bureau of Fisheries and Aquatic Resources (BFAR) Region 5
7. Cyrus Job P. Dela Cruz
   EMS I,
   CENRO Goa

Thematic Area 4. Education in STEAM and Culture

1. Dr. Minnie E. Talaguit
   Public Schools District Supervisor (PSDS),
   Department of Education
2. Atty. Louise Faye D. Enciso
   Attorney IV / Provincial Legal Officer,
   National Commission on Indigenous People (NCIP) Region 5
3. Wendell O. Baynas
   Admin Aide III,
   National Commission on Indigenous People (NCIP) Region 5
4. Guillermo C. Ortua
   President,
   Goa Campus Alumni Association
5. Marichu C. Borre
   Alumni Representative,
   Partido State University Federation of Alumni Association

Thematic Area 5. Terrestrial and Marine Resources, Economy, Biodiversity and Conservation

1. John Ray Cambaling
   SEMS,
   LGU Tinambac
2. Edwin San Juan
   OIC-AHRD/BO III,
   Partido Development Administration
3. Jaime A. Coralde
   Chairman, Sagñay Rigaton Agriculture Cooperative
4. Kenneth P. Zamudio
   President,
   Rotary Club of Goa Partido
5. Alvin B. Burac
   OIC – Fish Capture,
   Bureau of Fisheries and Aquatic Resources (BFAR) Region 5
6. Geraldine Consuelo
   Technical Staff – LHD,
   Bureau of Fisheries and Aquatic Resources (BFAR) Region 5
7. Cyrus Job P. Dela Cruz
   EMS I, CENRO Goa

Thematic Area 6. Smart Analytics and Engineering Innovation

1. Patrocinio N. Felizmenio
   Provincial S&T Director,
   Department of Science and Technology (DOST) Camarines Sur
2. Madel A. Alacar
   Trade-Industry Development Specialist,
   Department of Trade and Industry (DTI) Camarines Sur

Thematic Area 7. Gender and Development

1. Jocelyn R. Peralta
   Municipal Social Welfare Development (MSWD),
   LGU Goa
2. Jennelyn C. Lontoc
   Agricultural Technologist I,
   LGU San Jose
3. Aiza S. Romaraog
   PDO2,
   Department of Social Welfare and Development Region 5 (DSWD R5)

Partido State University

1. Dr. Raul G. Bradecina
   SUC President III
2. Prof. Patricia M. Candelaria
   VP for Research, Extension and Knowledge Management
3. Dr. Arnel B. Zarcedo
   VP for Administration and Finance
4. Ms. Luisa M. Lanciso
   Director, Research and Development
5. Ms. Rosemarie A. Tabardillo
   Director, Extension and Community Services
6. Dr. Mariel Estrella
   Director, Knowledge Management
7. Mr. Michael Vale
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8. Ms. Maria Cristina G. Gumba
   Director, Coastal and Wetland Ecosystems Center
9. Ms. Yumi Vivien de Luna
   Director, ParSU Statistics Center
10. Engr. Emelina Padayao
    Director, Water Resource Research Center/Dean, College of Engineering
11. Dr. Joni Neil Capucao
    Dean, College of Arts and Sciences
12. Ms. Kris T. Sarmiento
    Program Director
    Office Administration
13. Dr. Joan Monforte-Bedes
    Dean, College of Education
14. Dr. Shirlene Esplana
    Dean, Lagonoy Campus
15. Dr. Ricky Laureta
    Dean, San Jose Campus
16. Dr. Romeo Morcilla
    Dean, Tinambac Campus
17. Ms. Janeth Cledera
    Director, Quality Assurance
18. Engr. Kier Gasga
    Instructor I